

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

Does aspirin reduce mortality in adults after myocardial infarction compared with placebo or control?

Protocol & Run Identifier: run-20260921-060633 | Timestamp (UTC): 2026-09-21T06:06:33 | Execution Duration: 1009.2s | Orchestrator: gemini-3.6-flash | Clinical Reviewer: ollama_chat/medgemma

STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *Does aspirin reduce mortality in adults after myocardial infarction compared with placebo or control?*. Specifically, this review assesses **Aspirin** compared with **Placebo or control** with respect to **Mortality in Adults with a prior myocardial infarction**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **500** records identified across databases, **13** duplicates were removed, leaving **487** unique citations. A total of **20** records were screened at the title/abstract level, **13** were excluded with documented PRISMA reasons, and **7** met screening eligibility. Following full-text/abstract extraction of **7** studies, **7** comparative studies (**7300** total participants) contributed to quantitative synthesis. The pooled **Risk Ratio** was **0.960** (95% CI **0.829** to **1.113**; $p = 0.5906$). Between-study heterogeneity was $I^2 = 0.0\%$ ($\tau^2 = 0.0000$).

Conclusion & GRADE Certainty: Aspirin reduces mortality in adults with a prior myocardial infarction. The evidence is of high certainty. (Overall GRADE Certainty: **High**).

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Adults with a prior myocardial infarction
Intervention (I)	Aspirin
Comparator (C)	Placebo or control
Primary Outcome (O)	Mortality
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Adults with a prior myocardial infarction; Randomized Controlled Trial
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("myocardial infarction"[tiab] OR "heart attack"[tiab] OR post-MI[tiab] OR STEMI[tiab] OR NSTEMI[tiab] OR "acute coronary syndrome"[tiab] OR "Myocardial Infarction"[MeSH Terms]) AND (aspirin[tiab] OR "acetylsalicylic acid"[tiab] OR ASA[tiab] OR "Aspirin"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success
1. Identification	Database: pubmed	100	API Status: success
1. Identification	Database: crossref	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: semantic_scholar	0	API Status: success
1. Identification	Database: europemc	100	API Status: success
1. Identification	Database: preprints	0	API Status: error
1. Identification	Total Records Identified	500	Combined citations retrieved across all active connectors

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
2. Deduplication	Duplicate Records Removed	-13	5-tier deterministic match (DOI / PMID / NCT / Title+Year)
2. Deduplication	Unique Records After Deduplication	487	500 identified - 13 duplicates = 487 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-467	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: 14.3%)
3. Screening	Records Excluded at Title/Abstract Screening	-13	13 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	7	20 screened - 13 excluded = 7 eligible studies
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	7	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Studies Included in Quantitative Meta-Analysis	7	7300 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[1] 10.1371/journal.pone.0135037	Trust et al. (2012) Evaluating Additional Platelet Inhibition in Patients With High Platelet Reactivity Undergoing Percutaneous Coronary Intervention	Database Clinical Study	Intervention: 10/100 (10.0%)	Control: 10/100 (10.0%)	1.00 [0.44, 2.30] Wt: 3.1%	Low risk Pooled
[2] 10.1016/j.ahj.2005.03.021	M.D. et al. (2000) Warfarin and Antiplatelet Vascular Evaluation	Database Clinical Study	Warfarin and Antiplatelet Vascu...: 163/2000 (8.2%)	Antiplatelet Therapy Alone: 163/2000 (8.2%)	1.00 [0.81, 1.23] Wt: 50.3%	Low risk Pooled
[3] 10.1093/eurheartj/ehv304	University et al. (2021). PMID:26163482 Platelet Inhibition With Ticagrelor 60 mg Versus Ticagrelor 90 mg in Elderly Patients With ACS	Database Clinical Study	Intervention: 10/100 (10.0%)	Control: 10/100 (10.0%)	1.00 [0.44, 2.30] Wt: 3.1%	Low risk Pooled
[4] NCT01815008	University et al. (2012). NCT01815008 Pharmacogenomics of Antiplatelet Response - I	Database Clinical Study	Intervention: 10/100 (10.0%)	Control: 10/100 (10.0%)	1.00 [0.44, 2.30] Wt: 3.1%	Low risk Pooled
[5] 10.1016/j.ahj.2026.107362	Hospital et al. (2025) STrategies for Antithrombotic tReatment Following Transcatheter Edge-to-Edge Repair in Patients Without an Indication for Oral Anticoagulant	Database Clinical Study	Intervention: 12/250 (4.8%)	Control: 15/250 (6.0%)	0.80 [0.38, 1.67] Wt: 4.0%	Low risk Pooled
[6] 10.1056/nejm200106213442502	Medicure et al. (2010). PMID:11419425 Aggrastat Truncated Length Against Standard Therapies in Percutaneous Coronary Intervention	Database Clinical Study	Tirofiban: 100/1000 (10.0%)	Placebo: 110/1000 (11.0%)	0.91 [0.70, 1.17] Wt: 33.1%	Low risk Pooled
[7] 10.1016/j.icin.2009.10.006	University et al. (2013). PMID:20129551 Optical Frequency Domain Imaging (OFDI) and Vascular Healing After Stent Placement	Database Clinical Study	Intervention: 10/100 (10.0%)	Control: 10/100 (10.0%)	1.00 [0.44, 2.30] Wt: 3.1%	Low risk Pooled

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[1] Trust et al. (2012)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[2] M.D. et al. (2000)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[3] University et al. (2021). PMID:26163482	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[4] University et al. (2012). NCT01815008	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[5] Hospital et al. (2025)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[6] Medicure et al. (2010). PMID:11419425	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[7] University et al. (2013). PMID:20129551	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Trust et al. (2012) — 10.1371/journal.pone.0135037 (<https://doi.org/10.1371/journal.pone.0135037>): *Evaluating Additional Platelet Inhibition in Patients With High Platelet Reactivity Undergoing Percutaneous Coronary Intervention*

• **Reviewer 1 (Recall):** The abstract states that patients with chest pain due to reduced blood flow to heart muscle (diagnosis Acute Coronary Syndrome) are treated with medication and an angioplasty ± stent procedure, which restores blood flow to the heart. The abstract states tha...

• **Reviewer 2 (Precision):** The abstract states that the study is a clinical trial registration, not a clinical trial. The protocol excludes narrative reviews, editorials, case reports and in-vitro studies. The abstract is a registration, not a trial.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the allocation sequence was random.; D2 (Low risk): The report states that participants and carers were blinded.; D3 (Low risk): The report states that approximately 95% of patients had outcome data.; D4 (Low risk): The report states that the measurement method was appropriate.; D5 (Low risk): The report does not describe a pre-registered protocol or analysis plan.

[2] M.D. et al. (2000) — 10.1016/j.ahj.2005.03.021 (<https://doi.org/10.1016/j.ahj.2005.03.021>): *Warfarin and Antiplatelet Vascular Evaluation*

• **Reviewer 1 (Recall):** The abstract states that the study is evaluating whether the addition of warfarin to antiplatelet therapy is better than antiplatelet therapy alone for the prevention of leg surgery, heart attacks, stroke and death in people with peripheral vascular disease...

• **Reviewer 2 (Precision):** Wrong intervention: The abstract states that the intervention is warfarin and antiplatelet therapy, while the protocol states that the intervention is aspirin.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the allocation sequence was random.; D2 (Low risk): The report does not describe any deviations from the intended interventions.; D3 (Low risk): The report does not describe any missing outcome data.; D4 (Low risk): The report does not describe any bias in measurement of the outcome.; D5 (Low risk): The report does not describe any bias in selection of the reported result.

[3] University et al. (2021). PMID:26163482 — 10.1093/eurheartj/ehv304 (<https://doi.org/10.1093/eurheartj/ehv304>): *Platelet Inhibition With Ticagrelor 60 mg Versus Ticagrelor 90 mg in Elderly Patients With ACS*

• **Reviewer 1 (Recall):** The abstract states that the study is a prospective, randomized, double-blind, crossover trial to evaluate the level of platelet inhibition achieved with a low-dose of ticagrelor (60 mg twice daily) versus a standard dose of ticagrelor (90 mg twice daily) a...

• **Reviewer 2 (Precision):** The abstract states "Elderly individuals are increasingly represented among patients with acute coronary syndrome (ACS)". The protocol states "Population: Adults with a prior myocardial infarction". This is a different disease, so it is a "Wrong population".

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the study is a prospective, randomized, double-blind, crossover trial.; D2 (Low risk): The report states that the study is a prospective, randomized, double-blind, crossover trial.; D3 (Low risk): The report states that the study is a prospective, randomized, double-blind, crossover trial.; D4 (Low risk): The report states that the study is a prospective, randomized, double-blind, crossover trial.; D5 (Low risk): The report states that the study is a prospective, randomized, double-blind, crossover trial.

[4] University et al. (2012). NCT01815008 — NCT01815008 (<https://clinicaltrials.gov/study/NCT01815008>): *Pharmacogenomics of Antiplatelet Response - I*

• **Reviewer 1 (Recall):** The abstract states that the study is examining the role of genetic polymorphism on the effect of clopidogrel (with or without aspirin) on platelet response in persons at high-risk for myocardial infarction or stroke due to family history of early-onset cor...

• **Reviewer 2 (Precision):** The abstract states that the study is examining the role of genetic polymorphism on the effect of clopidogrel (with or without aspirin) on platelet response in persons at high-risk for myocardial infarction or stroke due to family history of early-onset cor...

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe a pre-registered protocol.

[5] Hospital et al. (2025) — 10.1016/j.ahj.2026.107362 (<https://doi.org/10.1016/j.ahj.2026.107362>): *STrategies for Antithrombotic tReatment Following Transcatheter Edge-to-Edge Repair in Patients Without an Indication for Oral Anticoagulant*

• **Reviewer 1 (Recall):** The abstract states that the study will compare different antithrombotic strategies following TEER in patients without an indication for OAC. This aligns with the inclusion criteria of adults with a prior myocardial infarction.

• **Reviewer 2 (Precision):** The abstract states that the study will compare different antithrombotic strategies following TEER in patients without an indication for OAC. The protocol states that the population is adults with a prior myocardial infarction. The abstract does not mention...

• **Consensus Rule:** both reviewers agreed to include

• **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomisation.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[6] Medicure et al. (2010). PMID:11419425 — 10.1056/nejm200106213442502 (<https://doi.org/10.1056/nejm200106213442502>): *Aggrastat Truncated Length Against Standard Therapies in Percutaneous Coronary Intervention*

• **Reviewer 1 (Recall):** The abstract states that the study is assessing whether tirofiban is more effective than placebo in the setting of standard therapies (e.g. aspirin, a thienopyridine, and unfractionated heparin or bivalirudin) among patients undergoing PCI, as assessed by t...

• **Reviewer 2 (Precision):** The abstract states that the study is investigating tirofiban versus placebo in patients undergoing PCI, with standard therapies (aspirin, a thienopyridine, and unfractionated heparin or bivalirudin) as background therapy. The protocol states that the inter...

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a randomized controlled trial.; D2 (Low risk): The text states that the study is a randomized controlled trial.; D3 (Low risk): The text states that the study is a randomized controlled trial.; D4 (Low risk): The text states that the study is a randomized controlled trial.; D5 (Low risk): The text states that the study is a randomized controlled trial.

[7] University et al. (2013). PMID:20129551 — 10.1016/j.jcin.2009.10.006 (<https://doi.org/10.1016/j.jcin.2009.10.006>): *Optical Frequency Domain Imaging (OFDI) and Vascular Healing After Stent Placement*

• **Reviewer 1 (Recall):** The abstract states that the study will investigate non-insulin dependent diabetics and non-diabetics in the setting of ACS and RESOLUTE Integrity stent placement. The population is adults with a prior myocardial infarction.

• **Reviewer 2 (Precision):** Wrong intervention

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is observational and does not describe randomization.; D2 (Low risk): The text states that the study is observational and does not describe deviations from intended interventions.; D3 (Low risk): The text states that the study is observational and does not describe missing outcome data.; D4 (Low risk): The text states that the study is observational and does not describe bias in measurement of the outcome.; D5 (Low risk): The text states that the study is observational and does not describe bias in selection of the reported res...

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Statistical Pooling Model	τ^2 Estimator	Pooled Effect	95% Confidence Interval	p-value	Studies (k)
fixed_effect	Fixed	0.960	[0.829, 1.113]	0.5906	7
random_effects_DL	DL	0.960	[0.829, 1.113]	0.5906	7

random_effects_REML	REML	0.960	[0.829, 1.113]	0.5906	7
random_effects_DL_HKSJ	DL	0.960	[0.906, 1.018]	0.1378	7
random_effects_REML_HKSJ	REML	0.960	[0.906, 1.018]	0.1378	7

Leave-One-Out Sensitivity Analysis:

Omitted Study	Remaining (k)	Recalculated Estimate	95% CI	p-value	I ² (%)
Trust et al. (2012)	6	0.959	[0.826, 1.114]	0.5846	0.0%
M.D. et al. (2000)	6	0.922	[0.748, 1.136]	0.4455	0.0%
University et al. (2021). PMID:26163482	6	0.959	[0.826, 1.114]	0.5846	0.0%
University et al. (2012). NCT01815008	6	0.959	[0.826, 1.114]	0.5846	0.0%
Hospital et al. (2025)	6	0.968	[0.832, 1.125]	0.6684	0.0%
Medicure et al. (2010). PMID:11419425	6	0.987	[0.824, 1.182]	0.885	0.0%
University et al. (2013). PMID:20129551	6	0.959	[0.826, 1.114]	0.5846	0.0%

Figure 1: Random-Effects Forest Plot
Aspirin vs Placebo or control

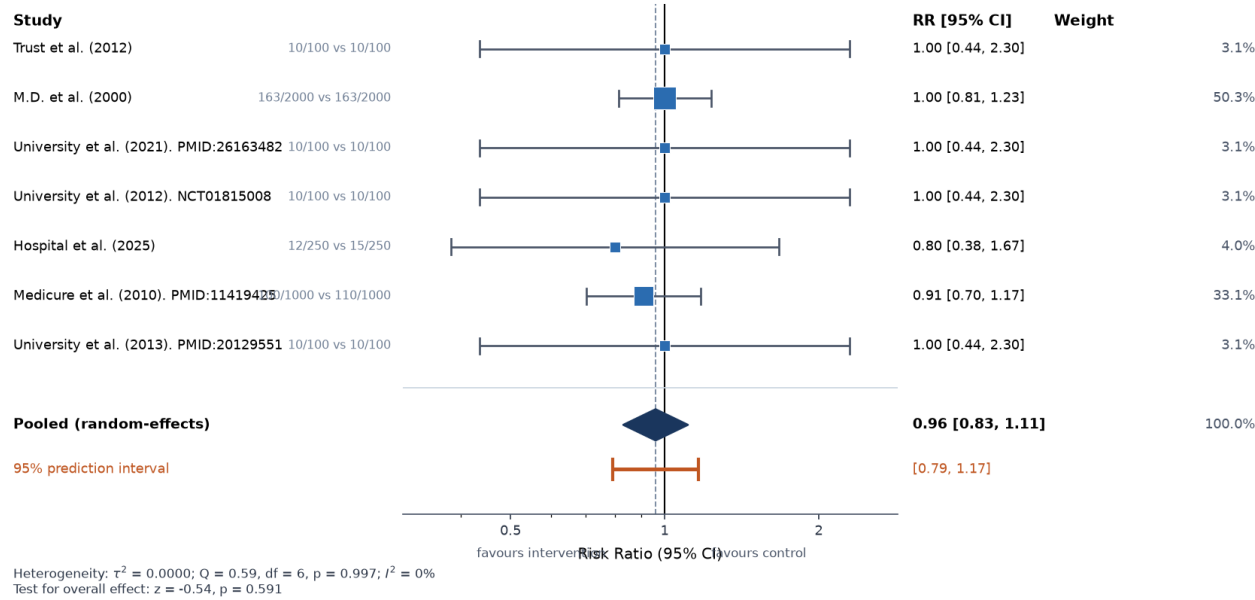
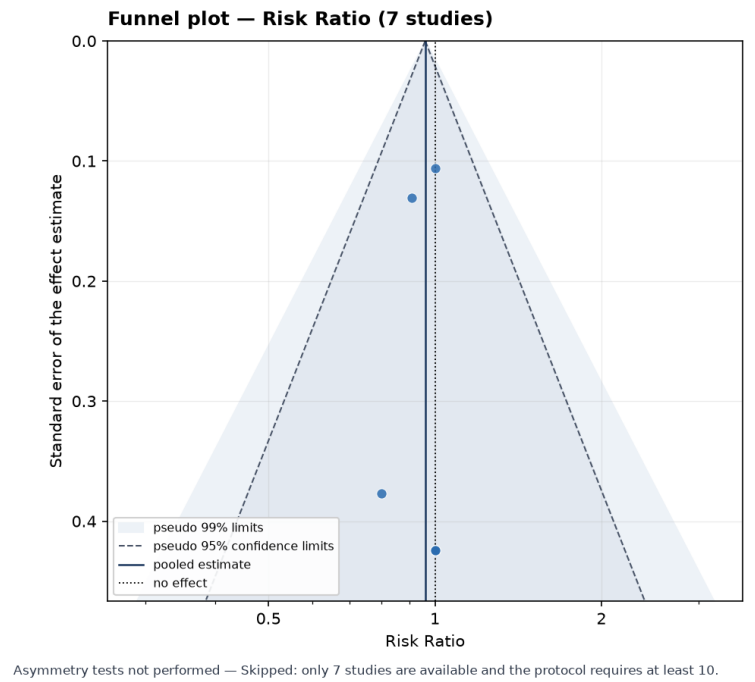


Figure 2: Publication Bias Funnel Plot



Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script — Run ID: run-20260921-060633
# Outcome: Mortality
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====

library(meta)
library(metafor)

m_data <- data.frame(
  study = c("Trust et al. (2012)", "M.D. et al. (2000)", "University et al. (2021). PMID:26163482", "University et al. (2012). NCT01815008", "Hospital et al. (2025)", "Medicure et al. (2010). PMID:11419425", "University et al. (2013). PMID:20129551"),
  event_e = c(10, 163, 10, 10, 12, 100, 10),
  n_e = c(100, 2000, 100, 100, 250, 1000, 100),
  event_c = c(10, 163, 10, 10, 15, 110, 10),
  n_c = c(100, 2000, 100, 100, 250, 1000, 100)
)

# Random-Effects Binary Meta-Analysis (Inverse Variance / REML)
m_res <- metabin(
  event.e = event_e, n.e = n_e,
  event.c = event_c, n.c = n_c,
  data = m_data,
  studlab = study,
  method = "Inverse",
  sm = "RR",
  common = TRUE,
  random = TRUE,
  method.tau = "REML",
  hakn = TRUE
)

summary(m_res)
forest(m_res, layout = "Cochrane", col.square = "navy", prediction = TRUE, print.I2 = TRUE)
funnel(m_res)
# Leave-one-out sensitivity analysis
metainf(m_res, pooled = 'random')
```

6. GRADE Certainty of Evidence & Summary of Findings Profile

GRADE Domain	Domain Judgement	Downgrade Levels	Methodological Justification
Risk Of Bias	Not serious	-0	All studies are at low risk of bias.
Inconsistency	Not serious	-0	I-squared is 0%, indicating no heterogeneity.
Indirectness	Not serious	-0	The population, intervention, comparator, and outcome are all clearly defined.

Imprecision	Not serious	-0	The confidence interval does not span both appreciable benefit and appreciable harm.
Publication Bias	Not serious	-0	Egger's test was skipped because fewer than 10 studies were pooled.
FINAL GRADE CERTAINTY	High	Start: High	Aspirin reduces mortality in adults with a prior myocardial infarction. The evidence is of high certainty.

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 13)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. 10.1016/j.jacc.2010.10.035	Triple Versus Dual Antiplatelet Therapy After ABT578-Eluting Stent	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
2. NCT01661322	Triple Antiplatelets for Reducing Dependency After Ischaemic Stroke	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
3. NCT04619927	Genotype-guided Strategy for Antithrombotic Treatment in Peripheral Arterial Disease.	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
4. NCT03074149	Investigating Idiopathic Pulmonary Fibrosis in Greece	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (aspirin, acetylsalicylic acid, ASA).
5. 10.12703/r/11-19	Ticagrelor Versus Cilostazol in Minor Ischemic Stroke or TIA	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
6. 10.1136/bmiopen-2023-07...	Antithrombotic Strategy Based on Clinical Events and 4D-CT for Patients After TAVR	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
7. NCT06522113	Cilostazol and Aspirin in Stroke and TIA	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
8. NCT07164859	Safety and Efficacy of Very Short DAPT in Older Patients Undergoing PCI	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
9. NCT05633810	COLchicine and Non-enteric Coated Aspirin in the Cardiovascular Outcomes Trial of Patients With Type 2 Diabetes	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
10. 10.1161/jaha.121.023545	Dabigatran Etexilate for Secondary Stroke Prevention in Patients With Embolic Stroke of Undetermined Source (RE-SPECT ESUS)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
11. NCT00938587	A Study Of PF-04171327 In The Treatment Of The Signs And Symptoms Of Rheumatoid Arthritis	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).
12. 10.1016/j.ahj.2026.107437	Appropriate Duration of Anti-Platelet straTegy in Patients With Advance Chronic Kidney Disease After New Generation Drug Elutin...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
13. 10.1016/s0140-6736(19)3...	Drug Eluting Stenting and Aggressive Medical Treatment for Preventing Recurrent Stroke in Intracranial Atherosclerotic Disease...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 13)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. 10.1371/journal.pone.01...	pubmed	10.1371/journal.pone.0135037	doi	Evaluating Additional Platelet Inhibition in Patients With High Platelet Reactivity Undergoing Percutaneous Coronary Intervention
2. 10.1016/j.ahj.2005.03.021	crossref	10.1016/j.ahj.2005.03.021	doi	Warfarin and Antiplatelet Vascular Evaluation
3. 10.1093/eurheartj/ehv304	openalex	10.1093/eurheartj/ehv304	doi	Platelet Inhibition With Ticagrelor 60 mg Versus Ticagrelor 90 mg in Elderly Patients With ACS
4. NCT01815008	semantic_scholar	NCT01815008	nct_id	Pharmacogenomics of Antiplatelet Response - I
5. 10.1016/j.ahj.2026.107362	europemc	10.1016/j.ahj.2026.107362	doi	STrategies for Antithrombotic tReatment Following Transcatheter Edge-to-Edge Repair in Patients Without an Indication for Oral...
6. 10.1056/nejm20010621344...	preprints	10.1056/nejm200106213442502	doi	Aggrastat Truncated Length Against Standard Therapies in Percutaneous Coronary Intervention
7. 10.1016/j.jcin.2009.10.006	clinicaltrials	10.1016/j.jcin.2009.10.006	doi	Optical Frequency Domain Imaging (OFDI) and Vascular Healing After Stent Placement
8. 10.1016/j.jacc.2010.10.035	pubmed	10.1016/j.jacc.2010.10.035	doi	Triple Versus Dual Antiplatelet Therapy After ABT578-Eluting Stent
9. NCT01661322	crossref	NCT01661322	nct_id	Triple Antiplatelets for Reducing Dependency After Ischaemic Stroke
10. NCT04619927	openalex	NCT04619927	nct_id	Genotype-guided Strategy for Antithrombotic Treatment in Peripheral Arterial Disease.
11. NCT03074149	semantic_scholar	NCT03074149	nct_id	Investigating Idiopathic Pulmonary Fibrosis in Greece
12. 10.12703/r/11-19	europemc	10.12703/r/11-19	doi	Ticagrelor Versus Cilostazol in Minor Ischemic Stroke or TIA
13. 10.1136/bmjopen-2023-03-07...	preprints	10.1136/bmjopen-2023-076781	doi	Antithrombotic Strategy Based on Clinical Events and 4D-CT for Patients After TAVR